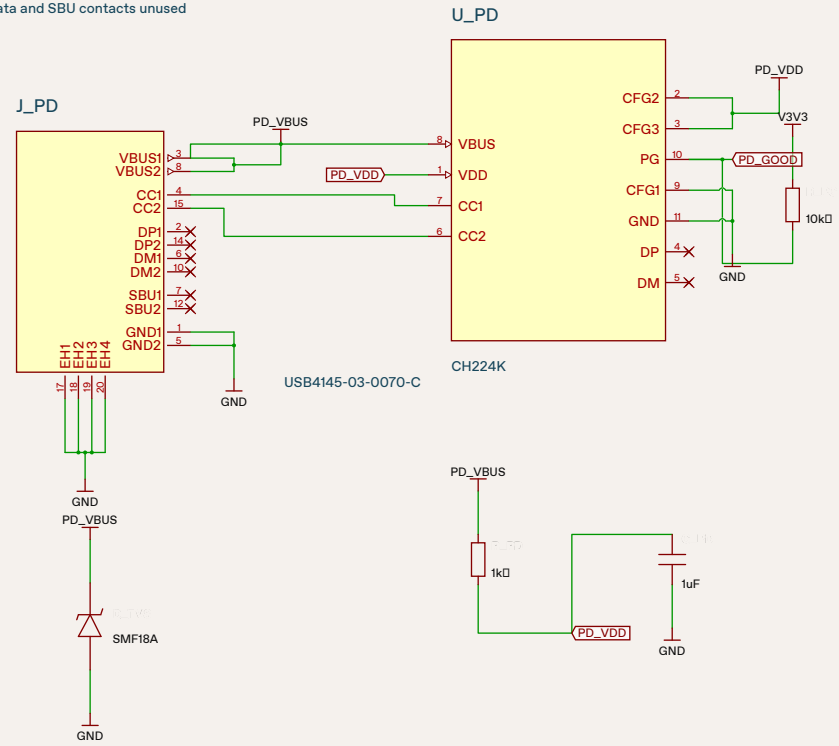
NEMA14 / CH32X035

Power & USB-PD

Motor supply negotiation, reverse-blocked logic power and 3.3 V conversion

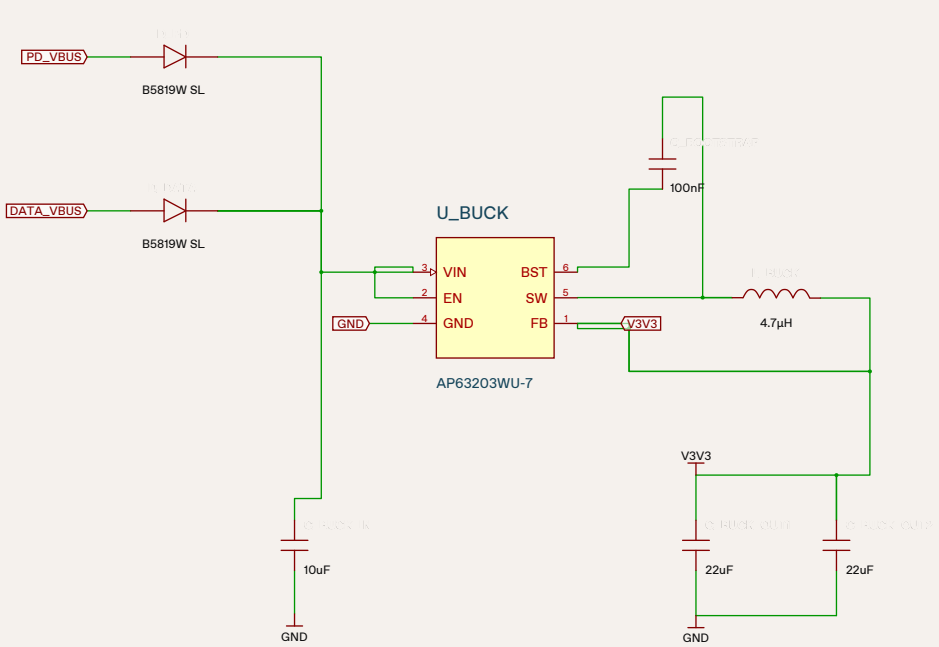
USB-C motor power

Fixed 15 V request; data and SBU contacts unused



3.3 V logic supply

AP63203 fixed-output buck; USB power is diode-isolated



PD_VBUS = motor rail DATA_VBUS = programming USB 5 V

VIN range and transients require bench qualification.

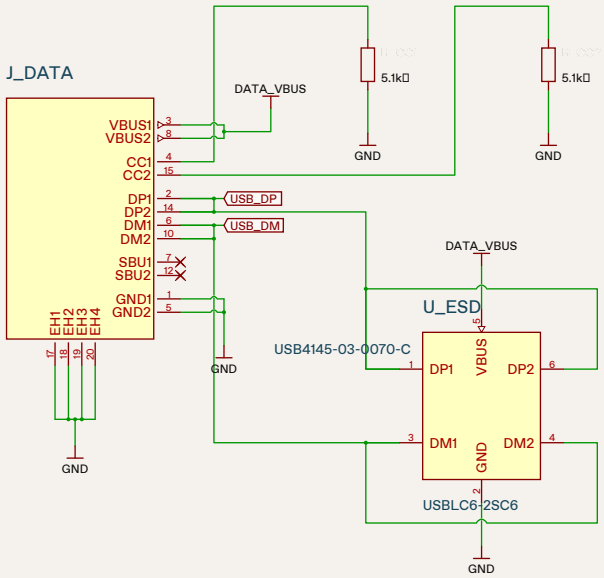
NEMA14 / CH32X035

Programming & USB

Programming USB-C, data-line protection and nonvolatile boot interfaces

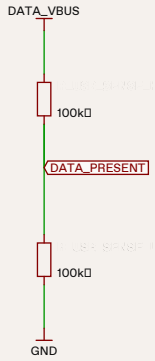
USB-C data port

Separate 5.1 kΩ CC pull-downs; duplicated D+/D− contacts



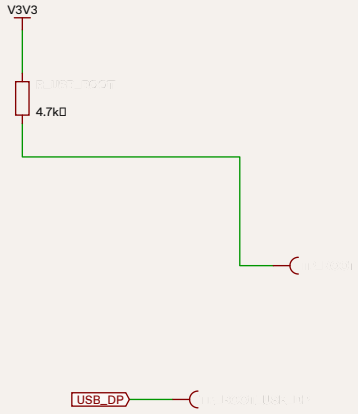
USB presence

Resistive divider to MCU; fixed 5 V input only



Boot & debug access

Boot contact and WCH DCK/DIO are retained



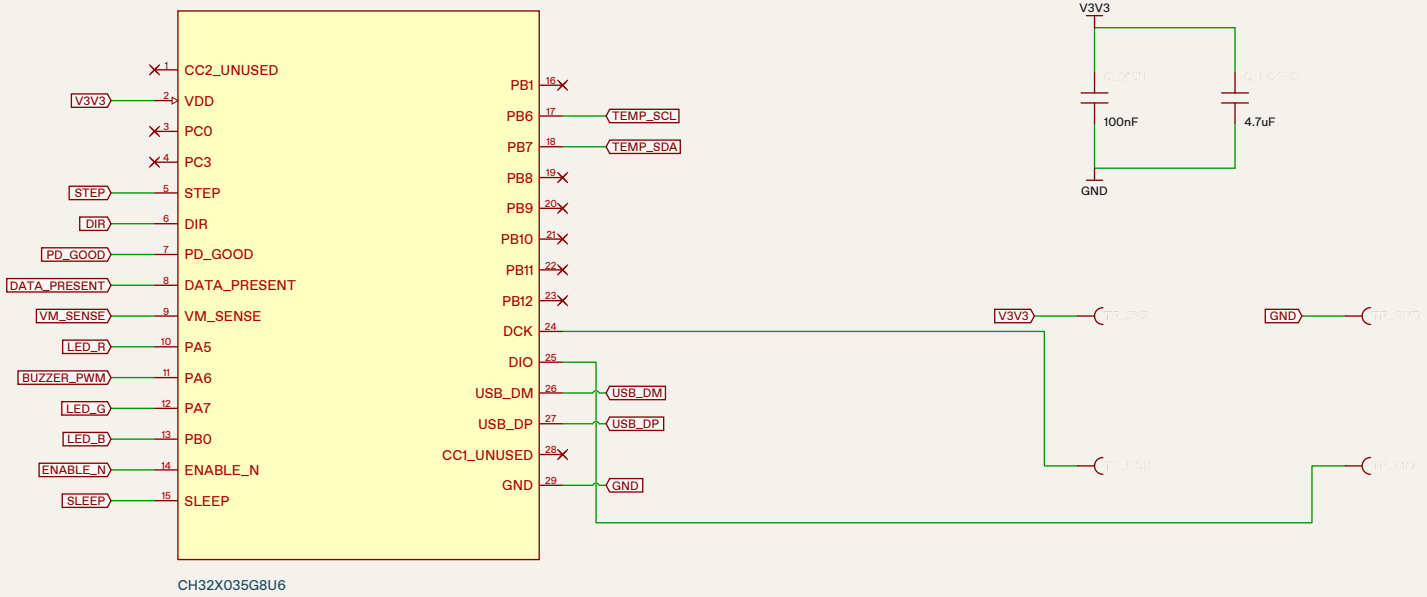
USB full-speed D+/D− route directly to CH32X035.

Debug interface is shown on Controller sheet 03.

Controller & clocks

MCU interfaces, reset/debug access, power decoupling and clock support

U_MCU



CH32X035G8U6

3.3 V bypass

Place bypass capacitors close to VDD

TMP102 uses software I²C: PB7 SDA / PB6 SCL.

Buzzer: PA6 TIM3_CH1. RGB: PA5 / PA7 / PB0.

Unused GPIO and USB-C CC pins are explicitly NC.

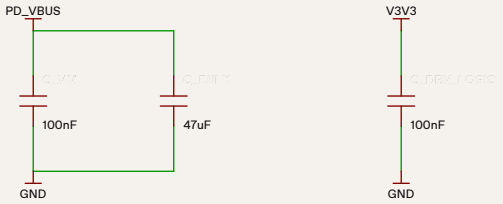
NEMA14 / CH32X035

Stepper motor driver

A4988 phase outputs, charge pump, current limit and motor-voltage measurement

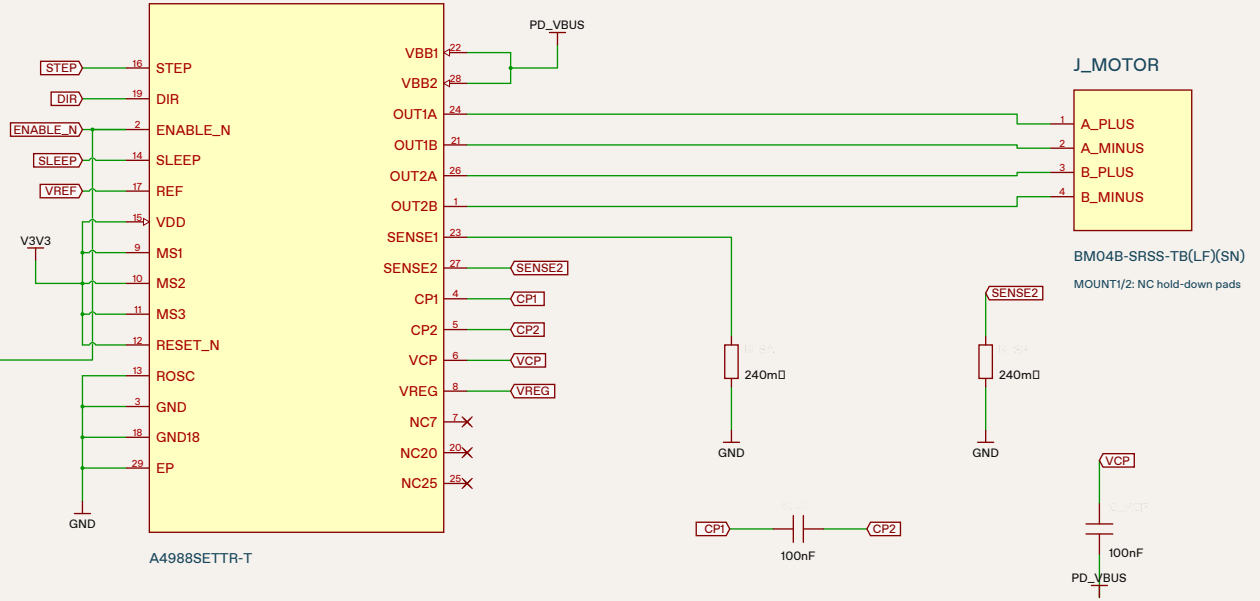
Driver supplies

Local bulk and high-frequency bypass



JST-SH pin order: 1 A+ 2 A− 3 B+ 4 B−

U_DRV



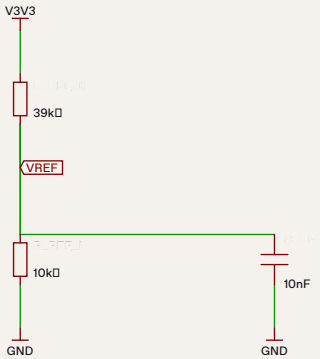
J_MOTOR

BM04B-SRSS-TB(LF)(SN)

MOUNT1/2: NC hold-down pads

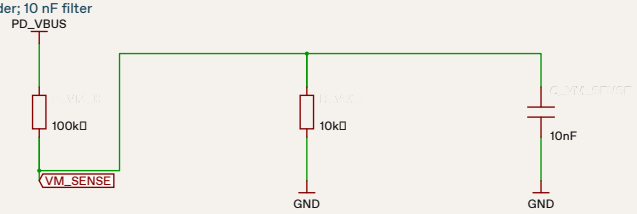
Current reference

VREF / (8 × RSENSE) = 0.351 A nominal



Motor voltage monitor

100 kΩ / 10 kΩ divider; 10 nF filter





AnasSarkiz / CH32X035 motor controller

35 × 35 mm | 4-layer PCB | Hardware prototype — assembly/firmware/bench qualification pending

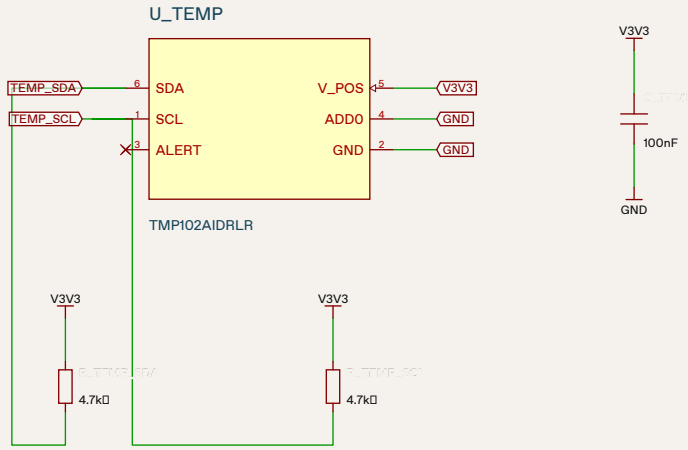
NEMA14 / CH32X035

Temperature & feedback

I²C temperature telemetry, RGB status outputs and PWM magnetic buzzer

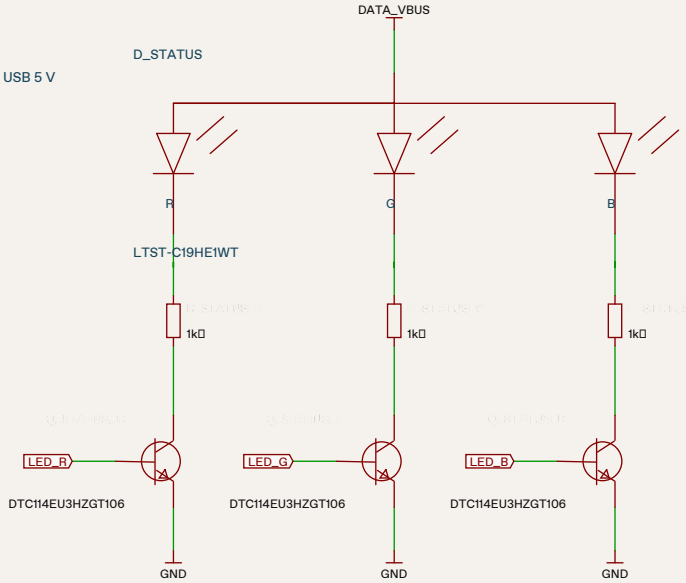
TMP102 temperature

ADD0 grounded: address 0x48; ALERT unused



RGB status indicator

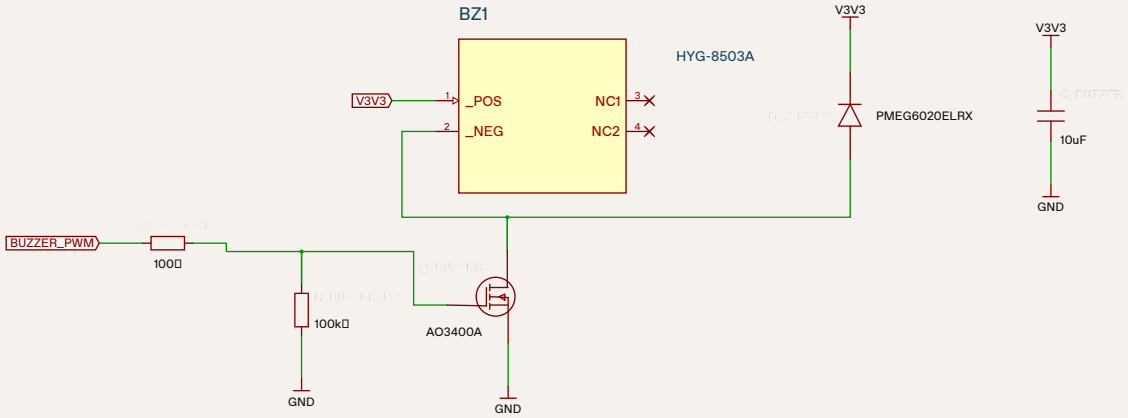
Common anode uses programming USB 5 V



DTC114E: NPN with internal input/bias resistors.

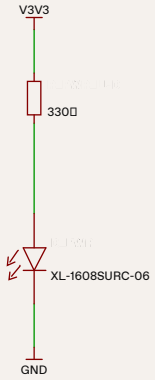
PWM buzzer

120 mA magnetic load; MOSFET drive and flyback diode



Power indication

3.3 V rail present; RGB/buzzer require firmware



AnasSarkiz / CH32X035 motor controller

35 × 35 mm | 4-layer PCB | Hardware prototype — assembly/firmware/bench qualification pending

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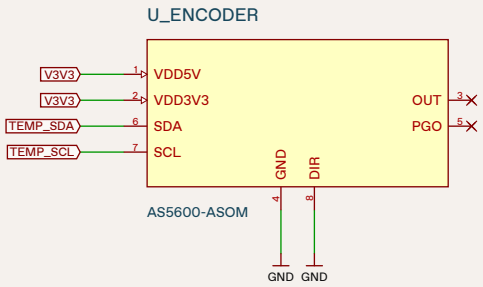
NEMA14 / CH32X035

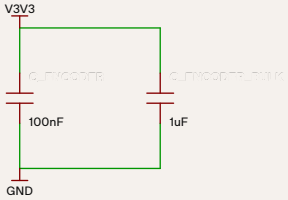
Optional encoder (DNP)

Bottom footprints only; AS5600 and both bypass capacitors omitted from factory assembly

Future magnetic shaft position

Nominal sensor centre (0, 0); rear shaft/magnet geometry remains unverified





U_ENCODER, C_ENCODER and C_ENCODER_BULK: bottom / DO NOT PLACE.

3.3 V mode: VDD5V + VDD3V3 tied together; DIR grounded; OUT/PGO NC.

Shared existing temperature I²C bus: AS5600 0x36 / TMP102 0x48.

Existing 4.7 k□ top-side pull-ups retained. Encoder firmware is required.

A diametrically magnetized shaft magnet and motor fit must be validated.